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5 / 5 submitted

Q1. Program based on Classes and Objects - Initializaing Data member using Objects directly

Score: 6.67 TC: 2/3 passed

CODE

```
import java.util.Scanner;
public class Student
{
    int rollNo;
    String name;
    double marks;

    void displayDetails()
    {
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }

    public static void main(String[] args)
    {
        Scanner sc= new Scanner(System.in);
        Student s= new Student();

        s.rollNo=sc.nextInt();
        sc.nextLine();
        s.name=sc.nextLine();
        s.marks=sc.nextDouble();

        s.displayDetails();
    }
}
```

INPUT

101
sripathi
90

OUTPUT

Roll No: 101
Name: sripathi
Marks: 90.0

Q2. Classes and Objects-Initialize Object Data Using Setter Methods

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Student
{
    private int rollNo;
    private String name;
    private double marks;

    public void setRollNo(int rollNo)
    {
        this.rollNo=rollNo;
    }
    public void setName(String name)
    {
        this.name=name;
    }
    public void setMarks(double marks)
    {
        this.marks=marks;
    }

    public void displayDetails()
    {
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args)
    {
        Scanner sc= new Scanner(System.in);

        Student s= new Student();

        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();

        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);

        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
101
sripathi
90
```

OUTPUT

```
Roll No: 101
Name: sripathi
Marks: 90.0
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Student
{
    int rollNo;
    String name;
    double marks;

    Student()
    {
        rollNo=101;
        name="Rahul";
        marks=85.5;
    }
    Student(int rollNo, String name, double marks)
    {
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }

    void displayDetails()
    {
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }

    public static void main(String[] args)
    {
        Scanner sc= new Scanner(System.in);

        Student s1= new Student();
        s1.displayDetails();

        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();

        Student s2= new Student(rollNo,name, marks);
        s2.displayDetails();

        sc.close();
    }
}
```

INPUT

```
102
Maram
100
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 102
Name: Maram
Marks: 100.0
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Employee
{
    int empId;
    String empName;
    double basicSalary;

    void getEmployeeDetails(int empId, String empName, double basicSalary)
    {
        this.empId=empId;
        this.empName=empName;
        this.basicSalary=basicSalary;
    }

    public static void main(String[] args)
    {
        Scanner sc= new Scanner(System.in);

        Salary s= new Salary();

        int id=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double salary=sc.nextDouble();

        s.getEmployeeDetails(id,name,salary);

        s.calculateSalary();
        s.displayDetails();

        sc.close();
    }
}

class Salary extends Employee
{
    double hra, da, netSalary;

    void calculateSalary()
    {
        hra=basicSalary * 0.20;
        da=basicSalary * 0.10;
        netSalary=basicSalary+hra+da;
    }

    void displayDetails()
    {
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Net Salary: "+netSalary);
    }
}
```

INPUT

```
101
Sripathi
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Employee
{
    int empId;
    String empName;

    void getEmployeeDetails(int empId, String empName)
    {
        this.empId=empId;
        this.empName=empName;
    }

    public static void main(String[] args)
    {
        Scanner sc= new Scanner(System.in);

        GrossSalary g= new GrossSalary();

        int id=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double salary=sc.nextDouble();

        g.getEmployeeDetails(id, name);
        g.getBasicSalary(salary);

        g.calculateSalary();
        g.displayDetails();

        sc.close();
    }
}

class Salary extends Employee
{
    double basicSalary;

    void getBasicSalary(double basicSalary)
    {
        this.basicSalary=basicSalary;
    }
}

class GrossSalary extends Salary
{
    double hra,da, grossSalary;

    void calculateSalary()
    {
        hra=basicSalary* 0.20;
        da=basicSalary * 0.10;
        grossSalary= basicSalary+hra+da;
    }

    void displayDetails()
    {
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
    }
}
```

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```
System.out.println("HRA: "+hra);
System.out.println("DA: "+da);
System.out.println("Gross Salary: "+grossSalary);
}

}
```

INPUT

```
101
Sripathi
500000
```

OUTPUT

```
Employee ID: 101
Employee Name: Sripathi
Basic Salary: 500000.0
HRA: 100000.0
Da: 50000.0
Gross Salary: 650000.0
```